Ex - 08
Data Visualization Using Tableau

AIM

To visualize data using Tableau by connecting to various data sources, creating bar charts, line charts, and pie charts, implementing calculated fields, and building interactive dashboards and stories using the Cardio dataset.

PROBLEM STATEMENT

To analyze and visualize the Cardio dataset using Tableau by connecting Excel, CSV, and SQL databases, creating basic charts, performing calculations, and developing interactive dashboards to present meaningful insights such as cardio cases by age group, blood pressure trends, and cholesterol distribution.

PROCEDURE

1. Open Tableau Desktop and explore the interface, including the Data Source page, worksheet, Rows shelf, Columns shelf, Marks card, and Show Me panel.
2. Connect to the Cardio dataset (CSV) by selecting Text file under Connect, choosing the file, and loading the required data.
3. Connect to a SQL database by selecting the required database connector, entering the server details, providing authentication if needed, and selecting the required tables.
4. Check the imported data, verify the data types, rename fields if necessary, and remove unwanted columns or records to prepare the dataset for visualization.
5. Create a bar chart by dragging Age Group to Columns and the count of id to Rows, then select Bar Chart from the Show Me panel and drag cardio to Color to compare patients with and without disease.
6. Create a line chart by dragging age to Columns and the average of ap_hi (systolic blood pressure) to Rows, then select Line from the Marks card to display the trend by age.
7. Create a pie chart by selecting Pie from the Marks card, dragging cholesterol to Color, and dragging the count of id to Angle to display the cholesterol level distribution.
8. Create a calculated field by selecting Analysis > Create Calculated Field, entering a formula, such as the BMI formula, and clicking OK to generate a new calculated field.
9. Build a dashboard by selecting New Dashboard, dragging the required worksheets into the dashboard, and arranging charts, titles, legends, and filters to create an interactive layout.
10. Create a story by selecting New Story, adding worksheets or dashboards as story points, providing captions, and saving the workbook for further analysis and presentation.

IMPLEMENTATION

1. Calculated Field - BMI

```
[weight] / (([height] / 100) ^ 2)
```

2. Calculated Field - BMI Category

```
IF [BMI] >= 30 THEN "Obese"
ELSEIF [BMI] >= 25 THEN "Overweight"
ELSE "Normal"
END
```

3. Calculated Field - Age Group

```
IF [age] >= 60 THEN "60+"
ELSEIF [age] >= 50 THEN "50-59"
ELSEIF [age] >= 40 THEN "40-49"
ELSE "30-39"
END
```

Dashboard components:

• Stacked bar chart - Patients by age group (colored by cardio)

• Line chart - Average systolic BP by age

• Pie chart - Cholesterol level distribution

• Bar chart - Cardio cases by BMI category

• Filter - Select gender, age group or cholesterol

• Story - Present visualizations in a sequence

OUTPUT

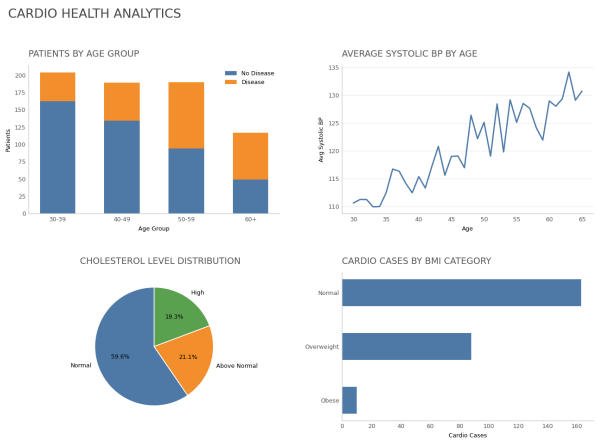


Fig: Cardio Health Analytics Dashboard (Tableau)

RESULT

Data visualization using Tableau was successfully studied and implemented on the Cardio dataset. Data was connected from different sources, bar charts, line charts, and pie charts were created, calculated fields were implemented, and interactive dashboards and stories were developed to analyze and present meaningful insights.